PROGRAMMING FUNDAMENTALS | C++

Lab 06

Class Work · Loops, Conditions & Logic Building

Lab 6 · Class Work

Factorial · Sum of Even Numbers · If-Else · Switch · Loops

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CLASS

BS 1st - D

SUBJECT

Programming Fundamentals (PF)

CONTENTS

- Q1 Factorial of a Number
- Q2 Sum of Even Numbers (0 to 100)
- Q3 · Q8 If-Else, Switch & Loop Practice Problems
- All questions include Code + Output

Question 1

```cpp
#include <iostream>
using namespace std;

int main() {
    for (int i = 5; i > 0; i -= 2) {
        cout << "Hello";
    }

    return 0;
}
```

Output:

Compile Result

```
HelloHelloHello
[Process completed - press Enter]
```

Lab 6 Class Work 5:

Write a program that takes the values a, b and c of a quadratic equation and prints both of its roots.

Code:

```cpp
#include <iostream>
#include <cmath>
using namespace std;
int main () {

    double x1, x2 ,a,b,c;
    cout << "Enter a,b and c." << endl;
    cin >> a >> b >> c;

    x1 =  (-b + sqrt(b*b - 4 * a * c)) / (2 * a);
    x2 =  (-b - sqrt(b*b - 4 * a * c)) / (2 * a);
    cout << "First Root is: " << x1 <<endl << "Second Root is: " << x2;
}
```

Output:

```
Enter a,b and c.
1
-2
1
First Root is: 1
Second Root is: 1
[Process completed - press Enter]
```

Question 3

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main() {
    int sum = 0;

    for (int i = 1; i <= 5; i++) {
        sum += i * 2;
    }

    cout << sum;

    return 0;
}
```

Output:

Compile Result

```
30
[Process completed - press Enter]
```

Question 2

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main() {
    for (int i = 2; i <= 10; i += 2) {
        cout << i << " ";
    }

    return 0;
}
```

Output:

Compile Result

```
2 4 6 8 10
[Process completed - press Enter]
```

Lab 6 Class Work 2:

Write a program that uses a loop to find the sum of all even numbers from 0 to 100 and prints the result.

Code:

```cpp
#include <iostream>
using namespace st
int main () {

    int sum = 0;
    for(int i=1; i<=100; i++)
    {
        if(i % 2 == 0) sum = sum + i;
    }
    cout << "The Sum of numbers from 0 to 100 is: " << sum;
    return 0;
}
```

Output:

```
The Sum of numbers from 0 to 100 is: 2550
[Process completed - press Enter]
```

Question 8

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main() {
    int x = 10;
    int y = 20;

    if (x < y)
        x = x + 5;
    else
        y = y + 5;

    cout << x << " " << y;

    return 0;
}
```

Output:

Compile Result

```
15 20
[Process completed - press Enter]
```

Lab 6 Class Work 1:

Write a program that takes a number from the user and prints its factorial.

Code:

```cpp
#include <iostream>
using namespace std;
int  main() {

    int i;
cout << "Enter a Number:" << endl;
cin >> i;
int fact = 1;
for(; i>=1;i--)
{
fact = fact * i;
}
cout << "Factorial of this number is: " << fact;
return 0;
}
```

Output:

```
Enter a Number:
5
Factorial of this number is: 120
[Process completed - press Enter]
```

Question 7

Coding C++ RUN MENU

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```cpp
#include <iostream>
using namespace std;

int main() {
    int x = 1;

    for (int i = 1; i <= 4; i++) {
        x *= 2;
    }

    cout << x;

    return 0;
}
```

Output:

Compile Result

```
16
[Process completed - press Enter]
```

Lab 6 Class Work 4:

Write a program that takes a number from 1 to 10 and prints it in Roman numerals using a switch statement. For any other number, print Invalid.

Code:

```cpp
#include <iostream>
using namespace std;
int  main() {

    int i;
    cout << "ENter a number from 1 to 10:" << endl;
    cin >> i;

    switch(i) {

    case 1:
    cout << "I";
    break;
    case 2:
    cout << "II";
    break;
    case 3:
    cout << "III";
    break;
    case 4:
    cout << "IV";
    break;
    case 5:
    cout << "V";
    break;
    case 6:
    cout << "VI";
    break;
    case 7:
    cout << "VII";
    break;
    case 8:
    cout << "VIII";
    break;
    case 9:
    cout << "IX";
    break;
    case 10:
    cout << "X";
    break;
    default:
    cout << "Invalid" << endl;
    }

    return 0;
}
```

Output:

```
Enter a number from 1 to 10:
7
VII
[Process completed - press Enter]
```

Lab 6 Class Work 3:

Write a program that takes three different numbers from the user and prints which one is the largest.

Code:

```cpp
#include <iostream>
using namespace std;
int main () {

    int a,b,c;
    cout << "Enter 3 numbers that are not equal: " << endl;
    cin >> a >>b >> c;

    if(a>b && a>c)
    cout << a << " is largest" << endl;
    else if(b>a && b>c)
    cout << b << " is largest" << endl;
    else cout << c << " is largest";

    return 0;
}
```

Output:

```
Enter 3 numbers that are not equal:
45
77
23
77 is largest

[Process completed - press Enter]
```

Question 4

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main() {
    int x = 3;

    for (int i = 1; i <= 3; i++) {
        x = x + i;
    }

    cout << x;

    return 0;
}
```

Output:

Compile Result

```
9
[Process completed - press Enter]
```

Question 6

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main() {
    int choice = 2;

    switch (choice) {
        case 1:
            cout << "One";

        case 2:
            cout << "Two";

        case 3:
            cout << "Three";
            break;

        default:
            cout << "Other";
    }

    return 0;
}
```

Output:

Compile Result

```
TwoThree
[Process completed - press Enter]
```

Question 5

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 7;

    if (n > 5) {
        if (n < 10)
            cout << "A";
        else
            cout << "B";
    }
    else {
        cout << "C";
    }

    return 0;
}
```

Output:

Compile Result

```
A
[Process completed - press Enter]
```